MIL-S-503B
28 APRIL 1988
SUPERSEDING
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# MILITARY SPECIFICATION

## SIRENS, ELECTRIC, 2 AND 5 HORSEPOWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers nondirectional electric motor driven sirens rated at 600 volts or less, complete with all controls and auxiliary equipment, suitable for mounting in the open on roofs or other supporting structures without additional protection from the weather.

1.2 Classification. Sirens should be of the following types, classes, and sizes, as specified (see 1.3, 3.5, and 6.2.1).

| TYPES | CLASSES | SIZES |
|---|---|---|
| Type I - Vertical | Class A - Uncoded | Size 2 - 2 Horsepower |
| Type II - Horizontal | Class B - Coded | Size 5 - 5 Horsepower |

1.3 Specification reference number. The specification reference number system should be definitive and identify each item of supply. The reference number shall be formatted using the types, classes, and sizes as follows (see 1.2, 3.9.1, and 6.2.1).

Definitive specification reference number: M 503 - X - X X

- M-prefix - an item defined by inch-pound (US) units
- Specification number
- Type
- Class
- Size

EXAMPLE: M503-I-B2; this identifies a vertical, coded, 2 horsepower siren.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350
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2.1.1 Specifications and standards. The following listed specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-T-60 | Tape, Packaging, Waterproof |
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-B-1055 | Barrier Material, Waterproofed, Flexible |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-V-173 | Varnish, Moisture and Fungus Resistant for Treatment of Communications, Electronic, and Associated Equipment |
| MIL-P-514 | Plates, Identification, Instruction and Marking |
| MIL-T-704 | Treatment and Painting of Material |
| MIL-S-3151 | Sound-Level Measuring and Analyzing Equipment |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Markings of U.S. Military Property |
| MIL-STD-143 | Specification and Standards, Order of Precedence for the selection of |
| MIL-STD-147 | Pallet Unit Loads |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-889 | Dissimilar Metals |
| MIL-STD-2073 | DOD Material Procedures for Development and Application of Packaging Requirements |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the document which is current on the date of the solicitation.



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM D-3951 | Packaging, Commercial |

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

| | |
|---|---|
| ICS | Industrial Controls and Systems |

(Application for copies of NEMA publications should be addressed to the National Electrical Manufacturers Association, 2101 "L" Street NW, Washington, DC 20037.)

UNDERWRITERS LABORATORIES INC (UL)

| | |
|---|---|
| UL-674 | Motors & Generators, Electric, for use in Hazardous Locations, Class I, Groups C & D; Class II, Groups E, F, & G. |

(Application for copies of UL publications should be addressed to Underwriters Laboratories Inc, 207 E. Ohio Street, Chicago, IL 60611.)

(Nongovernment standards and other publications are normally available from the organizations which prepares or which distributes the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Sirens shall be electric motor driven and shall be complete in all details. They shall be of rugged construction as to withstand vibration and exposure to severe weather conditions without excessive deterioration and shall be capable of operating after long periods of nonuse. Sirens shall be constructed so that dust, water, and condensation held within the unit will be at a minimum. Openings shall be screened with 0.250 inch galvanized steel hardware cloth to prevent the entrance of larger insects and animals. The siren shall be suitable for use on alternating current, voltage, frequency, and phase (single or three phase) as required.

3.2 First article. When specified in the contract or purchase order, one complete siren shall be subjected to first article inspection (see 4.4 and 6.2.1). The first article may be either a preproduction model or an initial production item which conforms to all requirements of this specification. In either case, the first article shall be identical to the production items in



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accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification. Unless otherwise specified (see 6.2.1), the first article shall remain at the manufacturer's plant as a production sample.

3.3 Selection of specifications and standards. Selection of specifications and standards for materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.4 Materials. Material shall be as specified herein. However, when a definitive material is not specified, a material shall be chosen of the quality used for the purpose in good commercial practice. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.4.1 Dissimilar metals. Dissimilar metals as defined by MIL-STD-889, shall not be used in intimate contact with each other unless adequately protected against galvanic corrosion.

3.4.2 Recycled and reclaimed material. Recycled or reclaimed material may be used provided all requirements of this specification are met and the material does not jeopardize the quality or life of the sirens.

3.4.3 Mounting hardware. Mounting hardware shall be selected from AN, NAS, MS or commercial standards to the maximum extent possible.

3.5 Construction. Construction of the siren and accessory equipment shall be in accordance with the best engineering practice. The overall equipment design and accessory installations shall permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. The siren, including the motor fan and stator, shall be suitably enclosed in a housing (see 3.7.2).

3.5.1 Types.

3.5.1.1 Type I - vertical. The siren rotor shall revolve about a vertical axis.

3.5.1.2 Type II - horizontal. The siren rotor shall revolve about a horizontal axis.

3.5.2 Sizes. The siren size shall be based on the rated horsepower output of the motor. The minimum sound level in decibels produced for each size siren shall be as specified herein when tested as required (see 4.7.2).

3.5.2.1 Size 2. The size 2 siren shall be driven by a motor that is rated for 2 horsepower output. The sound level produced by the siren, when tested as specified herein, shall not be less than 100 decibels when measured in a horizontal plane at 100 feet from the siren.



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3.5.2.2 Size 5. The size 5 siren shall be drive by a motor that is rated for 5 horsepower output. The sound level produced by the siren, when tested as specified herein, shall not be less than 1Ø5 decibels when measured in a horizontal plane at 1ØØ feet from the siren.

3.5.3 Classes.

3.5.3.1 Class A. The Class A siren is uncoded and shall operate only in response to a manually operated starter button.

3.5.3.2 Class B. The Class B siren is a coded siren and shall be arranged for the transmission of a manually controlled coded signal, a coded signal originating from a pull-lever type alarm box, or for the automatic transmission of one, two, three, or four predetermined, readily distinguishable signals. The coding shall not be accomplished by abrupt retardation of the siren motor speed.

3.6 Components.

3.6.1 Fan and stator. The fan and stator shall be made of aluminum. The fan and other rotating parts of the siren shall be in static and dynamic balance. The fan shall be fastened securely to the motor shaft and a key shall be used to prevent slipping. Baffles shall be provided within the housing or the design of the stator and housing shall be such that the input air is prevented from mixing with the output air.

3.6.2 Housing. The housing shall be constructed of zinc coated sheet metal or of .125 to .25Ø inch thick color impregnated fiberglass. All joints shall be riveted or welded securely, except that bolts shall be used to fasten portions made removable for inspection and maintenance of the interior mechanism. Sheet metal screws shall not be used. Fiberglass housings shall be drilled and shall be securely bolted to the supporting framework.

3.6.3 Base. A base shall be provided for mounting the siren on a flat horizontal surface. Anchor bolt holes shall be suitable for a minimum of .438 inch bolts.

3.6.4 Motors. Alternating current motors shall conform to the applicable requirements of UL-674; the horsepower rating shall be based on a 15 minute duty cycle, and a temperature rise of not more that 1Ø4°F (4Ø°C) above an ambient temperature of 122°F (5Ø°C). All motors shall be of the ball bearing type using prelubricated, environmentally sealed and shielded bearings. This arrangement shall eliminate the necessity of frequent maintenance. Motors shall be suitable for operation on the voltage, frequency, and phase (single or three phase), as specified (see 6.2.1).

3.6.5 Starters (control cabinets). Motor starters (control cabinets) shall be magnetic across the line type conforming to NEMA ICS. Thermal overload protection shall be provided and shall be arranged for automatic reset. Starters shall be of the proper size and suitable for the specific application. Unless otherwise specified (see 6.2.1), the enclosure shall be NEMA type 1, general purpose. When specified (see 6.2.1), the enclosure shall be NEMA type 3R, raintight type.



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3.6.6 Push-buttons. Push-buttons shall be of the maintained contact type, start-stop conforming to NEMA ICS. Unless otherwise specified (see 6.2.1), push-buttons shall be mounted on NEMA type 1 enclosure, general purpose. When specified (see 6.2.1), push-buttons shall be mounted on NEMA type 3R enclosure and shall be of the waterproof type.

3.6.7 Code transmitter. The controls for class B, coded siren shall include a code transmitter suitable for the transmission of coded signals as specified in 3.5.3.2. Code signals shall be as specified (see 6.2.1).

3.7 Operation.

3.7.1 Performance characteristics. When tested as specified (see 4.7.2), the sound produced by the siren shall start with a loud growl and shall rise in pitch to a shrill shriek, with 36Ø° horizontal sound radiation. The sound energy produced by the siren at maximum pitch shall be in the frequency range between 25Ø and 125Ø cycles per second. The time required for the siren to produce maximum pitch shall not exceed six (6) seconds from the instant the start of operation is initiated.

3.7.2 Sound level pattern. When tested as specified (see 4.7.2), the sound level pattern shall be such that the signal strength at any point, shall not be less than 1ØØ decibels for a size 2 siren or 1Ø5 decibels for a size 5 siren. Rotation of a siren to obtain a nondirectional sound level pattern shall not be permitted.

3.7.3 Environmental requirements.

3.7.3.1 Operating conditions. When tested as specified (see 4.7.1) the siren shall operate in any ambient temperature from -4Ø° Fahrenheit (F) (-4Ø° Celcuis (C)) to 125°F (51.7°C) at any possible relative humidity within that range.

3.7.3.2 Storage condition. When tested as specified (see 4.7.3.1) the siren shall not be damaged by storage at a low ambient temperature of -65°F (-53.9°C) for a period of twelve hours or a high ambient temperature of 155°F (68.3°C) for a period of four hours, at any possible relative humidity within the temperature range.

3.7.3.3 Rain resistance. When tested as specified (see 4.7.3.2) the siren shall not show any evidence of water penetration or water damage to operating or electrical parts of the siren.

3.7.3.4 Temperature-humidity resistance. When tested as specified (see 4.7.3.3) the siren shall not have an insulation resistance test potential of less than 5ØØ volts/Ø.1 megohm. The siren shall be adequately sealed to prevent moisture penetration into the electrical and operating parts that would impair the required operational performance.

3.7.4 Component fungus and moisture resistance. When specified (see 6.2.1), the electric circuitry, including all components and connections, shall be protected from the effects of fungus growth and moisture by an overall treatment



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with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.6.1), with 7 ±1 percent salicylanilide or 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish with the following exceptions:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed shall not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. Example: Current carrying contact surface, such as relay contact points.

When used, the varnish shall be applied by spray, brush, or a combination of both, to give a minimum dry film thickness of 0.001 inch to the component or element surface previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.7.4.1 Composition. Composition II shall be used only in the case where local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required.

3.8 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2.1), the unit surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2.1), the unit surfaces normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. In either case, the finish coat color shall be as specified (see 6.2.1).

3.8.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, all mill scale, products of corrosion, dirt, or other foreign substances which would inhibit proper coating adhesion.

3.8.2 Metal treatment. As soon as practical after cleaning and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating.

3.8.3 Painting. Painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practical after cleaning and treating. Painting shall be with manufacturer's current materials according to the manufacturer's current processes except that the total dry film thickness shall not be less than 2.5 mils.

3.9 Identification marking. Each siren shall be marked in accordance with MIL-STD-130. The marking shall be applied to each siren on identification plates made of corrosion resistant metal conforming to MIL-P-514. Plates shall be attached by means of screws, bolts, or rivets made of corrosion resistant metal in a location where the plates will be both visible and legible.



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3.9.1 Specification reference number. The specification reference number (see 1.3) shall be included as a part of the required identification marking.

3.10 Workmanship.

3.10.1 Steel fabrication. Steel used to fabricate the sirens and associated equipment shall be free of kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made by controlled means in order to insure uniformity of size and shape.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled; those through metal shall have all burrs removed, those through fiber glass shall be without any cracking or splitting of material and cleaned of all excess material residue. Washers or lockwashers shall be provided in accordance with good commercial practice with all bolts, nuts, and screws sufficiently tight to prevent removal without tools.

3.10.3 Riveted connections. Rivet holes shall be accurately punched or drilled; those through metal shall have all burrs removed, those through fiber glass shall be without any cracking or splitting of material and cleaned of all excess material residue. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the material.

3.10.4 Welding. Welding shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.10.5 Castings. Casting shall be sound and free from patching, misplaced coring, warping, or other defects which might render the castings unsound for use.

3.10.6 Paint. After each siren is painted, the paint shall be free from runs, sags, orange peel, or other visual defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all operational inspections shall be performed in accordance with the test conditions specified in paragraph 4.7.

4.4 First article inspection.

4.4.1 Examination. When specified, one sample siren of each type, class, and size shall be examined as specified in 4.6. Presence of one or more defects shall be cause for rejection.

4.4.2 Inspection. When specified, one sample siren of each type, class, and size shall be subjected to the inspections listed in 4.7. Failure of any inspection condition shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling from each lot for examination and inspection shall be in accordance with MIL-STD-105.

4.5.1.1 Inspection lot. All sirens of the same type, class, and size offered for delivery at one time shall be considered a lot for purposes of inspection.

4.5.2 Examination. The samples selected in accordance with 4.5.1 shall be examined for compliance with the requirements specified in section 3 of this specification.

4.5.3 Inspections. Quality conformance inspections shall consist of all the inspections specified in 4.6.1 and 4.7.3, based on inspection level II, MIL-STD-105. Presence of one major or two minor defects per 4.6.1 or failure of 4.7.3 or any combination thereof on four percent of the inspected sirens shall be cause for rejection.

4.6 Inspection procedure.



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4.6.1 Examination. The sirens shall be examined for the following defects:

| Major | Description | Paragraph |
|---|---|---|
| 101. | Workmanship not as specified. | 3.10 |
| 102. | Material not as specified. | 3.4 |
| 103. | Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other. | 3.4.1 |
| 104. | Construction not as specified. | 3.5 |
| 105. | Type not as specified. | 3.5.1 |
| 106. | Size not as specified. | 3.5.2 |
| 107. | Class not as specified. | 3.5.3 |
| 108. | Components missing. | 3.6 |
| 109. | Materials are not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environment. | 3.6 |
| 110. | Fan and stator not as specified. | 3.6.1 |
| 111. | Housing size not as specified. | 3.6.2 |
| 112. | Base/mounting capability not as specified. | 3.6.3 |
| 113. | Motor not as specified. | 3.6.4 |
| 114. | Starter/control cabinet not as specified. | 3.6.5 |
| 115. | Motor-control switches not as specified, when applicable. | 3.6.5 |
| 116. | Push-buttons not as specified. | 3.6.6 |
| 117. | Code transmitter not as specified. | 3.6.7 |
| 118. | Performance not as specified. | 3.7.1 |
| 119. | Sound pattern not as specified. | 3.7.2 |
| 120. | Fungus and moisture resistance treatment not as specified. | 3.7.4 |
| **Minor** | | |
| 201. | Mounting hardware missing or not as specified. | 3.4.3 |
| 202. | Treatment and painting not as specified. | 3.8 |
| 203. | Identification marking not as specified. | 3.9 |
| 204. | Technical publications missing or not as specified, when applicable. | |

4.7 Inspections. The first article shall be tested in accordance with 4.7.1, 4.7.2, and 4.7.3. Each quality conformance sample selected in accordance with 4.5 shall be tested in accordance with 4.7.3.

4.7.1 Operation. The siren shall be tested as specified in MIL-STD-810, method 501.2 (high temperature), procedure I, except that the temperature shall be 125°F (51.7°C). The siren shall then be tested as specified in MIL-STD-810, method 502.2 (low temperature), procedure I, except that the temperature shall be



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-40°F (-40°C). Failure of the siren to produce a sound starting with a heavy growl and rising in pitch to a shrill shriek at either temperature or any evidence of damage shall constitute failure of this test.

4.7.2 Sound level test. The siren shall be mounted approximately 20 feet above the ground at the center of a 200 foot diameter circle. The field this circle is located in shall be level and free from obstructions. The tests shall be made on a clear day, with a wind velocity of less than five miles per hour, and in an ambient temperature between 59° to 77°F (15° to 25°C). A sound level meter conforming to MIL-S-3151 shall be located at one of eight equally spaced predetermined points along the circle circumference with the microphone mounted five feet above the ground. With the microphone located on circumference of the test circle, diaphragm oriented directly toward the siren, the siren shall be adjusted ±3 feet for maximum sound level readings. The sound level test measurements shall then be made to determine the sound output and sound pattern of the siren. Each siren test shall commence at the zero degree azimuth, by producing three steady 15-second blasts and one steady 3-minute blast. The average sound level of the three 15-second blasts and the instantaneous sound level measured at 30-second intervals during the 3-minute blast shall be recorded. The average sound level of the three 15-second blasts shall constitute the siren sound output. The sound level of any steady blast shall not vary more than two (2) decibels. The sound pattern shall be determined by producing a 15-second blast at each of the eight predetermined points along the test circle and recording the average sound level.

4.7.3 Environmental.

4.7.3.1 Storage. The siren shall be placed in a temperature controlled chamber with the temperature stabilized at 155°F (68.3°C) and held for a period of four hours; immediately the temperature shall be lowered to -65°F (-53.9°C) and held for a period of twelve hours. The siren shall then be removed from the chamber, allowed to return to room temperature and then examined. Any evidence of damage (see 6.4) shall constitute failure of this test.

4.7.3.2 Rain inspection. The siren shall be mounted in a rain chamber to simulate actual installation conditions in accordance with the manufacturer's installation instructions. The temperature in the chamber shall be approximately 77°F (25°C) at the start of the test and no further regulation of temperature shall be made. A simulated rainfall of 4 inches ±1 inch per hour shall be produced by water spray nozzles of such design that water is emitted in the form of small droplets rather than a fine mist and at a rate equal to or exceeding 40 miles per hour. The temperature of the water shall be above 40°F (4.4°C). The spray nozzles shall be located so the water droplets impinge on the siren from all possible angles from the vertical to 45 degrees from the vertical. The water shall be dispersed as uniformly as possible over the entire area. During the last hour of the test, the siren shall be operated at the rated voltage to simulate actual operating conditions for a total operating time of not less than 15 minutes. Immediately after completion of the test, the siren shall be examined for evidence of water penetration or damage. Any evidence of water penetrating the enclosure so as to come in contact with operating or electrical parts, or any evidence of damage as a result of the test, shall be cause for rejection of the siren.



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4.7.3.3 Humidity inspection. The siren having passed the rain test, shall have the power supply removed; reinstalled in a test chamber to simulate actual installation conditions to be used for the humidity test. The test chamber shall be capable of being sealed, having a heat source so arranged that the heat does not directly impinge upon the siren being tested. The velocity of air in the test chamber throughout the test are shall not exceed 15Ø feet per minute. The siren shall not cover more than 5Ø percent of the volume and floor area of the test chamber. Distilled or demineralized water having a hydrogenation concentration (ph) value of between 5.4 and 7.5 at 77°F (25°C), shall be used to produce and maintain a relative humidity of 85 ±3 percent at an ambient chamber temperature of 86°F ±3° (3Ø°C ±1.6°). The chamber shall be sealed with the humidity and temperature test conditions stabilized and maintained for a period of not less than 24 hours. Within one hour after completion of the temperature humidity exposure and without removal of moisture from the siren, the insulation resistance of the electrical circuits shall be measured using a megohm meter with a test potential of not less than 5ØØ volts. A visual inspection of the siren for evidence of damage and moisture penetration into electrical and operating parts that would impair performance shall then be accomplished. Insulation resistance of less than Ø.1 megohm, damage, or moisture penetration that would impair performance shall be cause for rejection.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING.

5.1 Preservation and packaging. The preservation and packaging shall be level A, C, or commercial as specified (see 6.2.1).

5.1.1 Level A. All openings into the siren housing shall be sealed with tape conforming to PPP-T-6Ø, class 1, or barrier material conforming to PPP-B-1Ø55, class E-1 or E-2, secured with tape specified herein. Each control cabinet shall be preserved in accordance with MIL-P-116, method IIB.

5.1.2 Level C. Each complete siren shall be preserved and packaged in accordance with the applicable requirements of MIL-STD-2Ø73.

5.1.3 Commercial. Each complete siren shall be preserved and packaged in accordance with ASTM D-3951.

5.2 Packing. The packing shall be level A, B, C, or commercial, as specified (see 6.2.1).

5.2.1 Level A. Each complete siren shall be packed in a close-fitting box conforming to PPP-B-6Ø1, overseas type. The contents shall be cushioned, blocked, and braced, or otherwise secured to prevent movement inside the container or damage to the contents.

5.2.2 Level B. Each complete siren shall be packed as specified for level A, except that the boxes shall conform to PPP-B-6Ø1, domestic type.



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5.2.3 Level C. Each complete siren shall be packed in accordance with the applicable requirements of MIL-STD-2Ø73.

5.2.4 Commercial. Each complete siren shall be packed in accordance with the applicable requirements of ASTM D-3951.

5.3 Marking. Interior packages and shipping containers shall be legibly marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

5.4 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities, to the greatest possible extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. Notes.

6.1 Intended use. The sirens are intended for use as fire alarms or other type warning devices or systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Specification reference number for siren(see 1.3).

c. When first article required (see 3.2).

d. When first article is not the last siren delivered on contract (see 3.2).

e. When NEMA type 2R enclosure is required (see 3.6.5).

f. When push-buttons are required in NEMA type 3R enclosure (see 3.6.6).

g. Type of code signal required (see 3.6.7).

h. When cleaning, treatment, and painting in accordance with MIL-T-7Ø4 is required (see 3.8).

i. Color of finish coat required (see 3.8.3).

j. Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

k. When palletization is required (see 5.4).



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6.3 First Article. When a first article inspection is required, the siren should be a sample selected as specified in the contract or purchase order (see 3.2). The first article should consist of the number of sirens specified in the contract or purchase order. The contracting officer will include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For purposes of defining various types of defects or terminology which is referred to in this specification, the following definitions will apply:

a. Damage - Damage is corrosion, deformation, loosening, breakage, or any condition resulting in malfunctioning of the siren, component, or part.

b. Patching - Patching is repairs to any material used in the construction of the sirens.

c. Peel - Peel is the visually obvious condition of the paint or treatment cracking or separating from the base material to which applied.

d. Run - Run is a visual flow of the paint or treatment from a sag.

e. Sags - Sag is the visual condition caused by the excessive application of paint or treatment to the base material.

f. Warping - Warp is the visual condition of the base material being distorted from its true form during manufacture which is not a part of the planned completed product.

6.5 Cross-reference. Size 3 horsepower of JAN-S-503, dated 29 SEP 47, has been deleted because the increase in sound output (decibels) over the size 2 horsepower in negligible.

6.6 Subject term (key word) listing.

Audible siren
Decibels
Fire alarm
Siren
Sound patterns
Warning device



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6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 99

Review activities:
DLA - GS

User activities:
Army - ME, MI
Navy - MC
DLA - CS

Preparing activity:
Air Force - 82

Agent activity:
Air Force - 99

(Project 6350-0106)



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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-503B | Sirens, Electric, 2 and 5 Horsepower |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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